

Preliminary Exam in Analysis, January 2026

Problem 1. Let

$$f(x, y) = \sum_{n=1}^{\infty} \frac{x}{x^2 + yn^2}, \quad y > 0.$$

Show that for each $y > 0$, the limit

$$g(y) := \lim_{x \rightarrow +\infty} f(x, y)$$

exists. Evaluate the limit function g , and show that $f(x, y)$ converges to $g(y)$ uniformly for $y \in (0, \infty)$ as $x \rightarrow +\infty$.

Hint: Estimate the sum by an integral as in the Integral Test.

Problem 2. Assume that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is Lipschitz continuous and $L : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear. Let

$$A = \{(x_1, \dots, x_n) \in \mathbb{R}^n : |x_n| \geq x_1^2 + \dots + x_{n-1}^2\}.$$

Prove that if

$$\lim_{A \ni x \rightarrow 0} \frac{f(x) - f(0) - Lx}{|x|} = 0,$$

then f is differentiable at $0 \in \mathbb{R}^n$.

Hint: Think about the problem geometrically before doing any calculations.

Problem 3. Prove the Banach Contraction Principle: If (X, d) is a complete metric space and $T : X \rightarrow X$ satisfies

$$d(T(x), T(y)) \leq kd(x, y) \quad \text{for some } 0 \leq k < 1 \text{ and all } x, y \in X,$$

then there is a unique $x_* \in X$ such that $T(x_*) = x_*$.

Problem 4. Suppose that f is a positive smooth function on $\mathbb{R}^2$, such that for each fixed $x \in \mathbb{R}$, $f(x, y)$ is an increasing function in y . Let h be a smooth function on $\mathbb{R}$ such that $h(0) = \iint_{[0,1]^2} f$. Show that there is an open interval I containing 0, and a smooth function $g : I \rightarrow \mathbb{R}$ such that

$$h(x) = \int_0^1 \int_{g(x)u/2}^{\exp(g(x)u)} f(u, v) dv du, \quad \text{for each } x \in I.$$

Problem 5. Let $\mathcal{M}$ be a surface in $\mathbb{R}^3$ defined by the equation $z = f(x, y)$, where $f \in C^2(\mathbb{R}^2)$. Prove that if

$$f_{xy}^2(0, 0) < f_{xx}(0, 0)f_{yy}(0, 0),$$

then there is a neighborhood U of $p = (0, 0, f(0, 0))$ in $\mathbb{R}^3$ such that if T is the tangent plane to $\mathcal{M}$ at p , then $\mathcal{M} \cap T \cap U = \{p\}$.

Problem 6. Let C be a simple, closed, smooth curve passing through the origin in $\mathbb{R}^2$ and parametrized by smooth $\mathbf{r}(t)$ for $t \in [0, 1]$. For each $n \geq 3$, define C_n to be the curve in $\mathbb{R}^2$ parametrized by $\mathbf{r}(t)$ for $t \in [\frac{1}{n}, 1 - \frac{1}{n}]$. Let $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a smooth vector function such that $|\operatorname{curl} \mathbf{F}(x, y)| \leq e^{-x^2 - y^2}$ for each $(x, y) \in \mathbb{R}^2$. Show that there exists $M > 0$ such that for all $n \geq 3$ we have

$$\left| \int_{C_n} \mathbf{F} \cdot \mathbf{T} ds \right| \leq \pi(1 - e^{-(\operatorname{diam} C)^2}) + \frac{M}{n}.$$